PROBLEM DA10

Two-Belt Viscous Pump

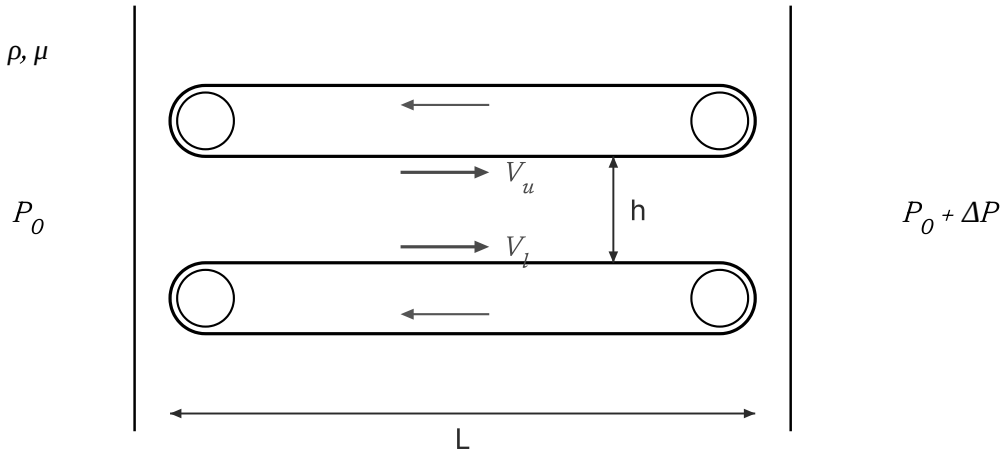
Original statement · DA10.pdf

Π theorem | Re ≪ 1 | Calibration table

OFFICIAL PROBLEM STATEMENT

PROBLEM 3 (50 minutes) The system sketched in the figure is designed to steadily pump fluid of constant density ρ and constant viscosity μ from a reservoir, where the modified pressure has a constant value P_0, to another one at pressure $P_0 + \Delta P$ ($\Delta P > 0$). To that end, a pair of parallel belts of length L, separated by a distance $h \ll L$, are driven at constant velocities V_l and V_u, dragging the liquid towards the large pressure reservoir.

In the forthcoming analysis, the liquid flow can be assumed steady and two-dimensional (spanwise length much larger than h). The most important parameters of the system are the liquid volume flux **per unit spanwise length**, q, the power needed to drive the system **per unit spanwise length**, $\dot{W}$, and the pump efficiency, $\eta = q\Delta P/\dot{W}$.



Statement figure (redrawn from DA10.pdf): two parallel belts of length L, a distance h ≪ L apart, driven at V_u and V_l, drag liquid of density ρ and viscosity μ from a reservoir at reduced pressure P_0 to one at P_0 + ΔP

1. (*3 points*). Use dimensional analysis to simplify the dependence of q, $\dot{W}$ and η, as functions of an appropriately chosen set of parameters governing the operation of the pump.

2. (*2 points*). Simplify the resulting dimensionless functions when the inertia of the fluid can be neglected compared to the viscous forces. Note that, in this limit, the fluid density plays a negligible role. In the remainder of the problem, assume that these conditions are satisfied.

A pump with $h = 1$ cm and $L = 20$ cm is calibrated using glycerol, of density $\rho_g = 1260$ kg/m^3 and viscosity $\mu_g = 1.41$ kg/(m s), using an overpressure $\Delta P_g = 1$ kPa. During the calibration process, the lower belt is kept at rest, $V_{l,g} = 0$, while the upper belt is driven at several velocities $V_{u,g}$, as indicated in the table below, where the corresponding values of q_g, $\dot{W}_g$ and η_g are also shown.

$V_{u,g}$ **(cm/s)**	q_g **(cm^2/s)**	$\dot{W}_g$ **(Watt/m)**	η_g
10	2	0.8	0.26
15	4.5	1.4	0.33
20	7	2.1	0.33
25	9.5	3	0.32
30	12	4	0.3
35	14.5	5.2	0.28

The **same pump** is used to deliver a silicone oil, of density $\rho_{oil} = 1100$ kg m^{-3} and viscosity $\mu_{oil} = 5$ kg/(m s), with an overpressure $\Delta P_{oil} = 2$ kPa.

3. (*2 points*). Determine the velocities $V_{l,oil}$ and $V_{u,oil}$ for which the results obtained in the calibration tests can be used to predict the pumping behavior with the silicone oil.

4. (*2 points*). Calculate the corresponding values of q_{oil}, $\dot{W}_{oil}$ and η_{oil}.

5. (*1 point*). Discuss the hypothesis of negligible fluid inertia.

`Source: DA10.pdf (page 1: statement; page 2: handwritten key). Notes.pdf, Chapter 6: Π theorem, Eq. 6.25; selection of parameters, p. 68; limits, Eq. 6.55; physical similarity, Eqs. 6.57-6.59.`

Worked solution

Hypotheses, Data and Dimensional Table

- Steady, two-dimensional flow of a liquid with constant ρ and μ; gravity is absorbed in the modified pressure, so only the difference ΔP enters (Notes.pdf, p. 66).
- All quantities are per unit spanwise length: $[q] = L^2T^{-1}$, $[\dot{W}] = ML^2T^{-3}/L = MLT^{-3}$.
- Recomputing $\eta = q\Delta P/\dot{W}$ from the table: row 1 gives $2 \times 10^{-4} \times 10^3/0.8 = 0.25$ (printed 0.26) and row 2 gives $4.5 \times 10^{-4} \times 10^3/1.4 = 0.32$ (printed 0.33); rows 3 to 6 agree with the printed values (0.333, 0.317, 0.300, 0.279).

Parameter	**Dimensions**
q	L^2T^{-1}

Parameter	Dimensions
$\dot{W}$	MLT^{-3}
η	1
ρ	ML^{-3}
μ	$ML^{-1}T^{-1}$
ΔP	$ML^{-1}T^{-2}$
V_u, V_l	LT^{-1}
L, h	L

Application of the Π Theorem

$$q = f_1(\rho, \mu, \Delta P, V_u, V_l, L, h), \qquad \dot{W} = f_2(\dots), \qquad \eta = f_3(\dots)$$

$n = 7$. The key takes (μ, V_u, h), natural for a viscous pump. Rows $\mu(-1, 1, -1)$, $V_u(1, 0, -1)$, $h(1, 0, 0)$ in (L, M, T); only μ has mass, so expanding along the M column, $\det = -1 \cdot (1 \cdot 0 - (-1) \cdot 1) = -1 \neq 0$: $k = 3$, and $n - k = 4$ groups remain.

Flux. $[q] = [\mu]^a[V_u]^b[h]^c$: $M:\ 0 = a$; $T:\ -1 = -b$; $L:\ 2 = b + c$; so the scale is $V_u h$.

Power. $[\dot{W}] = [\mu]^a[V_u]^b[h]^c$: $M:\ 1 = a$; $T:\ -3 = -a - b, b = 2$; $L:\ 1 = -a + b + c$, $c = 0$; so the scale is μV_u^2.

Density. $[\rho] = [\mu]^a[V_u]^b[h]^c$: $M:\ 1 = a$; $T:\ 0 = -1 - b, b = -1$; $L:\ -3 = -1 - 1 + c$, $c = -1$: the group $\rho V_u h/\mu$, a Reynolds number.

Pressure. $[\Delta P] = [\mu]^a[V_u]^b[h]^c$: $M:\ 1 = a$; $T:\ -2 = -1 - b, b = 1$; $L:\ -1 = -1 + 1 + c, c = -1$: the group $h\Delta P/(\mu V_u)$.

Belts and geometry. V_l/V_u and L/h.

$$\frac{q}{V_u h} = \Pi_q\left(\frac{\rho V_u h}{\mu}, \frac{h\Delta P}{\mu V_u}, \frac{V_l}{V_u}, \frac{L}{h}\right), \qquad \frac{\dot{W}}{\mu V_u^2} = \Pi_W(\dots), \quad \eta = \eta(\dots)$$

with the same four arguments.

Negligible Inertia

When inertia is negligible against viscous forces, the density drops out ($\rho V_u h/\mu \to 0$; Notes.pdf, Eq. 6.55):

$$\frac{q}{V_u h} = \Pi_q\left(\frac{h\Delta P}{\mu V_u}, \frac{V_l}{V_u}, \frac{L}{h}\right), \quad \frac{\dot{W}}{\mu V_u^2} = \Pi_W\left(\frac{h\Delta P}{\mu V_u}, \frac{V_l}{V_u}, \frac{L}{h}\right), \quad \eta = \eta\left(\frac{h\Delta P}{\mu V_u}, \frac{V_l}{V_u}, \frac{L}{h}\right)$$

Physical Similarity: from Glycerol to Silicone Oil

The pump is the same, so L/h is the same. The other two groups must be equal.

Lower belt.

$$\frac{V_{l,oil}}{V_{u,oil}} = \frac{V_{l,g}}{V_{u,g}} = 0 \implies V_{l,oil} = 0$$

Upper belt. The same h cancels:

$$\frac{h\Delta P_{oil}}{\mu_{oil} V_{u,oil}} = \frac{h\Delta P_g}{\mu_g V_{u,g}} \implies \frac{V_{u,oil}}{V_{u,g}} = \frac{\mu_g}{\mu_{oil}}\frac{\Delta P_{oil}}{\Delta P_g} = \frac{1.41}{5} \times \frac{2}{1} = 0.564$$

Oil Performance

With all groups equal, the dimensionless outputs are equal:

$$\frac{q_{oil}}{V_{u,oil} h} = \frac{q_g}{V_{u,g} h} \implies \frac{q_{oil}}{q_g} = \frac{V_{u,oil}}{V_{u,g}} = 0.564$$

$$\frac{\dot W_{oil}}{\mu_{oil} V_{u,oil}^2} = \frac{\dot W_g}{\mu_g V_{u,g}^2} \;\Longrightarrow\; \frac{\dot W_{oil}}{\dot W_g} = \frac{\mu_{oil}}{\mu_g}\left(\frac{V_{u,oil}}{V_{u,g}}\right)^2 = \frac{5}{1.41} \times 0.564^2 = 3.546 \times 0.3181 = 1.128$$

$$\eta_{oil} = \eta_g$$

Check: $\eta_{oil}/\eta_g = (q_{oil}/q_g)(\Delta P_{oil}/\Delta P_g)/(\dot W_{oil}/\dot W_g) = 0.564 \times 2/1.128 = 1$. The η_{oil} column copies the printed η_g, as the key does; from $q\Delta P/\dot W$ the first two rows would read 0.25 and 0.32.

$V_{u,g}$ (cm/s)	$V_{u,oil}$ (cm/s)	q_{oil} (cm^2/s)	$\dot W_{oil}$ (W/m)	η_{oil}
10	5.64	1.128	0.902	0.26
15	8.46	2.538	1.579	0.33
20	11.28	3.948	2.369	0.33
25	14.10	5.358	3.384	0.32
30	16.92	6.768	4.512	0.30
35	19.74	8.178	5.866	0.28

Negligible Inertia: Discussion

The plain Reynolds number of the calibration is not small: $\rho_g V_{u,g} h/\mu_g = 1260 \times 0.10 \times 0.01/1.41 = 0.89$ at the lowest speed and 3.1 at the highest. The key compares inertia and viscosity in the **slender** channel ($h \ll L$), where the convective term is smaller than the viscous one by the extra factor h/L, so the relevant number is the reduced Reynolds number $(\rho V_u h/\mu)(h/L)$. This criterion belongs to the slender viscous flows of Chapter 7 of the notes (an **extension** here). With $h/L = 1/20$:

$$\text{glycerol: } 0.89 \times \frac{1}{20} = 0.045 \;\le\; \frac{\rho_g V_{u,g} h}{\mu_g}\,\frac{h}{L} \;\le\; 3.13 \times \frac{1}{20} = 0.156$$

$$\text{oil: } \frac{1100 \times 0.0564 \times 0.01}{5} \times \frac{1}{20} = 0.0062 \;\le\; \frac{\rho_{oil} V_{u,oil} h}{\mu_{oil}} \frac{h}{L} \;\le\; \frac{1100 \times 0.1974 \times 0.01}{5} \times \frac{1}{20} = 0.022$$

Both are small, more so for the oil, so neglecting inertia is a good approximation, and the calibration can be transferred to the oil.

Answers and Checks

- Reduced laws: $q/(V_u h), \dot{W}/(\mu V_u^2)$ and η are functions of $\rho V_u h/\mu, h\Delta P/(\mu V_u), V_l/V_u, L/h$; without inertia, of the last three only.
- Similar operation with oil: $V_{l,oil} = 0, V_{u,oil} = 0.564\, V_{u,g}$.
- Oil performance: $q_{oil} = 0.564\, q_g, \dot{W}_{oil} = 1.128\, \dot{W}_g, \eta_{oil} = \eta_g$ (table above).
- Inertia: reduced Reynolds number 0.045-0.156 (glycerol) and 0.006-0.022 (oil).
- Dimensions: μV_u^2 is $ML^{-1}T^{-1} \cdot L^2T^{-2} = MLT^{-3}$, a power per unit length; $h\Delta P/(\mu V_u)$ is $L \cdot ML^{-1}T^{-2}/(ML^{-1}T^{-1} \cdot LT^{-1}) = 1$.
- Check against the official key: DA10.pdf gives $V_{l,oil} = 0, V_{u,oil}/V_{u,g} = 0.564, q_{oil}/q_g = 0.564, \dot{W}_{oil}/\dot{W}_g = 1.128$ and the same table, with the inertia ranges 0.045-0.156 and 0.0062-0.022.